



Northern University

Of Business and Technology Khulna

Course Title: Mobile Computing Lab

Course Code: CSE 4204

Feature Compilation Report

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Enterprise Nexus – Multi Agent AI Business Automation OS with Security Intelligence

CSE 4204 - Mobile Computing Lab | Team CSE4204-8B-T04

1. Introduction

By Week 09, the Enterprise Nexus team has completed the major foundation of the project - backend, frontend, database, and AI integration (delivered in Week 08). This report documents the Week 09 milestone: bringing these components together into one integrated, demonstrable system, replacing dummy data with real system data, resolving integration issues, and preparing the project for the formal testing phase.

2. Team Information

Name	Student ID	Role
Nazmus Sakib	11220320888	Team Leader & AI Integration Lead
Shoeb Shariar Mashuk	11220320878	Backend Developer
Most Sumiya Sanjida	11220320874	Frontend Developer - 1
Sabrina Ibrahim	11220320895	Database Manager & Frontend Developer - 2

3. Summary of Completed Features

All core modules defined in the approved proposal and SRS are implemented end-to-end - backend API, database persistence, AI analysis (where applicable), and a connected frontend page.

Module	Key Capabilities	Status
Authentication	Register, login, JWT refresh, logout, forgot/reset password, RBAC	Completed
HR	CV upload & AI screening, candidate list/detail, stats, HR agent chat	Completed
Finance	Expense records, invoice & bulk upload, anomaly detection, finance agent	Completed
Support	Tickets, sentiment/urgency analysis, escalate/resolve, RAG replies	Completed

Module	Key Capabilities	Status
Analytics	KPI snapshot, AI reports, charts	Completed
Executive	Multi-agent orchestration, daily briefing, PDF export, chat	Completed
Security Intelligence	Audit log viewer, security stats, threat monitoring	Completed
Knowledge Base / RAG	Document upload, embeddings, semantic search	Completed
Admin	User & role management, knowledge management	Completed
Notifications	In-app notifications, preferences, real-time updates	Completed

4. Partially Completed Features

As of Week 09, no major feature from the approved proposal is left partially implemented. All ten core modules listed above are fully functional end-to-end (frontend, backend, database, and AI where applicable).

5. Remaining Features

No major feature from the approved proposal remains unimplemented. Remaining work for the upcoming weeks is limited to non-functional polish - further UI/UX refinement, expanded automated test coverage, and production deployment hardening - rather than missing functionality.

Planned Enhancement - HR Agent Public Job Posting (n8n)

As a planned extension beyond the current scope, the team intends to make the HR Agent more functional by building a new public-facing job posting workflow, orchestrated primarily through n8n:

- One-click job posting - a single action from the HR Agent publishes a job opening.
- Automatic posting to a Telegram bot to broadcast the opening to candidates.
- A public apply link/page where candidates can submit applications without needing an account.
- Custom weighted requirements - each job requirement can be assigned a configurable weight for AI scoring.
- A configurable score threshold that determines automatic shortlisting.

This feature is not yet started and is planned for a later development cycle; it is listed here for transparency on project direction.

6. Frontend–Backend Integration Status

All major frontend components communicate correctly with the backend:

- API requests/responses verified across all modules (HR, Finance, Support, Analytics, Executive, Admin, Security).
- JWT-based authentication is enforced on protected routes via middleware.
- Database operations (create, read, update) are confirmed working with Row-Level Security enforcing per-user data isolation.
- Form submissions (CV upload, expense import, ticket creation, etc.) are wired to real backend endpoints - no hardcoded or static data remains where backend functionality is available.
- Data retrieval and updates use SWR (stale-while-revalidate) caching on the frontend to keep views in sync without redundant fetches.
- Centralized error handling (ErrorBoundary, standardized API error responses) surfaces understandable messages instead of raw technical errors.

7. AI Integration Status

The AI functionality delivered in Week 08 has been verified end-to-end:

- AI is connected to the actual application workflow (HR screening, finance anomaly detection, support triage, analytics reporting, executive briefing) - not a standalone demo.
- Users access every AI feature through the frontend (HR Agent, Finance Agent, Support Agent, Analytics Agent, Executive Agent pages).
- AI requests are routed through the backend (BaseAgent tool-calling layer) before reaching Google Gemini.
- AI responses are validated against a strict JSON schema and displayed correctly in the corresponding UI.
- API errors are handled gracefully - a hardened client retries transient failures and marks records `ai_status: "pending"/"failed"` instead of fabricating output.
- API credentials (Gemini key, Supabase service key) are stored server-side as environment variables and are never exposed to the client.
- AI functionality provides measurable value: CV scoring, anomaly flags, ticket sentiment/urgency, and cross-module executive briefings.

8. Authentication Status

The complete authentication workflow has been implemented and verified:

- Registration - creates a user with a bcrypt-hashed password and default "employee" role.
- Login - validates credentials, issues short-lived access and long-lived refresh JWTs as httpOnly cookies.
- Logout - revokes the stored refresh-token session server-side.
- Password handling - bcrypt hashing (cost factor 12) on register, login, and password update.
- Forgot / reset password - single-use, time-limited (1 hour) reset tokens; all existing sessions are invalidated on reset.

- Session management - refresh tokens are hashed and stored per-session in user_sessions, rotated on every refresh.
- Protected routes - auth middleware validates the access token on every protected request.
- Role-based access (RBAC) - role_id / roles relation gates Admin and Security Intelligence features.

9. Major Problems Encountered & Solutions Implemented

Area	Problem Encountered	Solution Implemented
AI output	Occasional non-JSON model responses broke downstream parsing	Switched to Gemini native JSON mode with an explicit response schema
Auth	Refresh token was not revoked on logout	Refresh tokens are now hashed, stored in user_sessions, and explicitly invalidated on logout
Finance	Bulk upload rejected some valid spreadsheets	Hardened file parsing and validation for CSV/Excel imports
Support	AI ticket replies were not grounded in company policy	Wired the retrieve_knowledge RAG tool into the reply-drafting step
UI	Layout broke at tablet-width breakpoints	Adjusted the responsive grid and sidebar collapse behaviour

10. Changes from Original Plan

Change 1 - Real-time mechanism

What was changed: The Week 07 roadmap originally planned Socket.io for real-time dashboard and activity updates.

Why it was changed: The project already depends on Supabase (PostgreSQL) for persistence; adding a separate Socket.io server would duplicate infrastructure and require managing a second stateful process in deployment.

New approach: Supabase Realtime (Postgres change-data-capture channels) is used instead, giving live dashboard refresh and an agent-activity feed with no additional server to operate.

Change 2 - Multi-agent orchestration framework

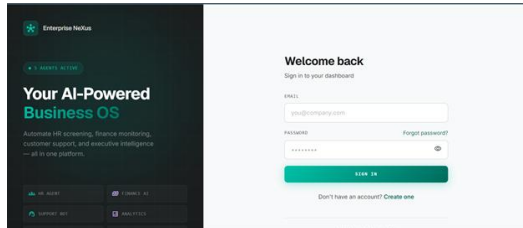
What was changed: The proposal referenced CrewAI for multi-agent orchestration.

Why it was changed: CrewAI's opinionated orchestration model added overhead that did not fit the project's simpler delegate-and-synthesize pattern between five agents.

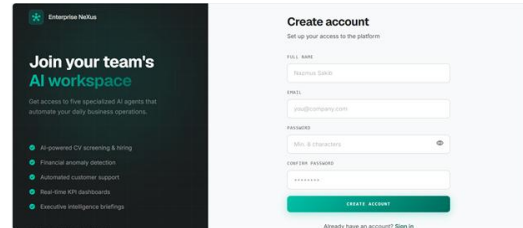
New approach: A custom BaseAgent tool-calling foundation built directly on LangChain.js (@langchain/google-genai) is used, with a dedicated Executive orchestrator that delegates to the four specialist agents and merges their output.

11. Current Project Screenshots

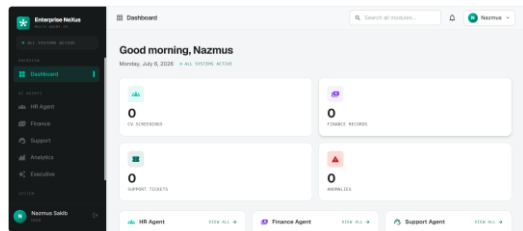
Representative screenshots of the current, integrated system:



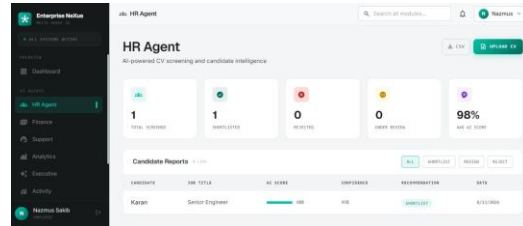
Login Page



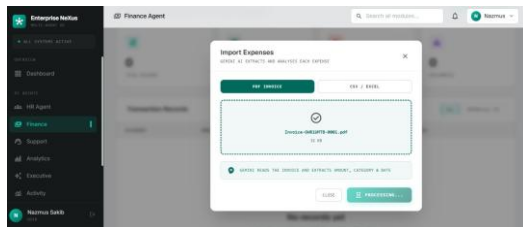
Registration Page



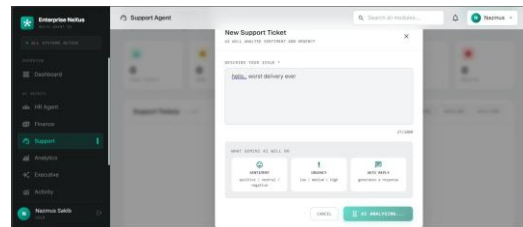
Main Dashboard



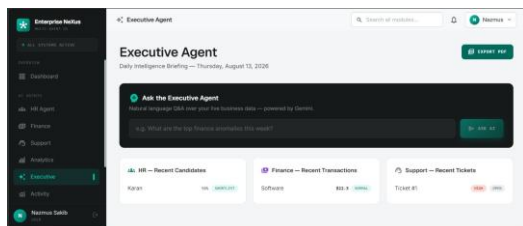
HR Agent - AI CV Screening



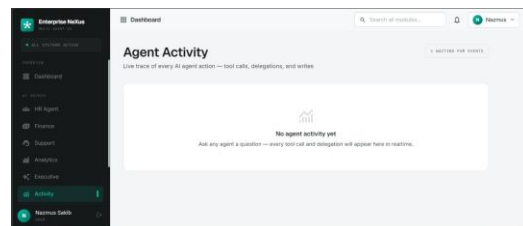
Finance Agent - Invoice Import



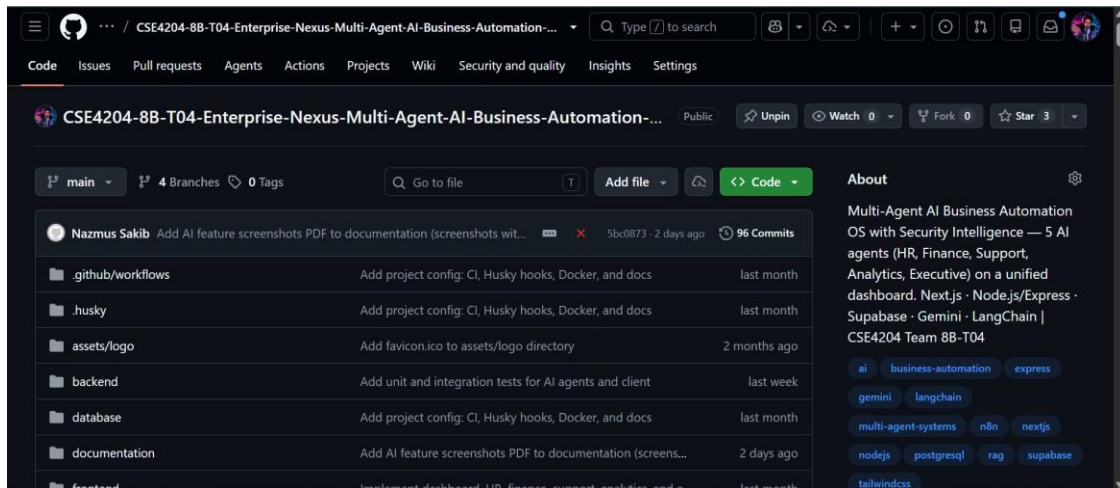
Support Agent - New Ticket (AI Sentiment/Urgency)



Executive Agent - Ask AI



Agent Activity - Live Trace Log



frontend	Implement dashboard, HR, finance, support, analytics, and a...	last month
postman	Add database schema (11 tables) and Postman API collectio...	2 months ago
screenshots	Add new AI feature screenshots and remove outdated imag...	5 days ago
supabase/migrations	Add pgvector migration for storing document embeddings	last week
.gitignore	Add light theme error pages and assets	2 months ago
.prettiignore	Add project config: CI, Husky hooks, Docker, and docs	last month
DEPLOYMENT.md	Add project config: CI, Husky hooks, Docker, and docs	last month
LICENSE	Add MIT License to the project	2 months ago
README.md	Update README.md to reflect new repository structure and...	5 days ago
docker-compose.yml	Add project config: CI, Husky hooks, Docker, and docs	last month
package.json	Add project config: CI, Husky hooks, Docker, and docs	last month
start.bat	Add project config: CI, Husky hooks, Docker, and docs	last month
stop.bat	Add project config: CI, Husky hooks, Docker, and docs	last month

tailwindcss

Readme

MIT license

Activity

3 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 4

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GitHub Repository & Recent Development Activity

12. GitHub Repository Link

Repository: <https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence>

13. Feature Completion Checklist

Feature	Status	Notes / Planned Completion (Week)
User Authentication (register/login/RBAC)	Completed	-
HR Agent - AI CV Screening	Completed	-

Feature	Status	Notes / Planned Completion (Week)
Finance Agent - Anomaly Detection & Invoice Import	Completed	-
Support Agent - Ticket Triage & RAG Replies	Completed	-
Analytics Agent - AI Reports	Completed	-
Executive Agent - Cross-module Orchestration	Completed	-
Security Intelligence Module	Completed	-
Knowledge Base / Semantic Search (RAG)	Completed	-
Admin Panel	Completed	-
Notifications	Completed	-
Automated Test Suite Expansion	In Progress	Planned for Week 10 (Testing & Debugging)
Production Deployment Hardening	In Progress	Planned for Week 11 (Deployment)
HR Agent - Public Job Posting Automation (n8n)	Not Started	Planned for Week 10-11